

Attempt all the questions:

GROUP 'B' [8×5=40]

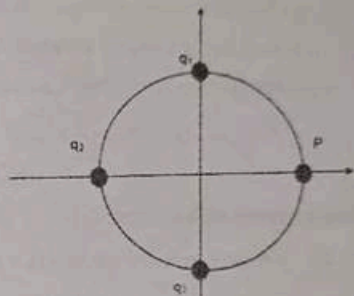
Short answer questions:

12. a) What are dimensional and non-dimensional variable? Give examples of each. [2]
- b) Check the correctness of the physical relation $Y = \frac{M g L}{\pi r^2 l}$ where Y = Young's modulus Mg is weight applied on wire, L is length of wire, l is increase in length of wire, and r is radius of wire. [2]
- c) How many times the units of energy be increased if each of the units of mass, length and time is increased by two times. [1]
13. a) Two people are pulling a heavy cart from the same point but in slightly different directions. One pulls with a force of magnitude A , the other with force B , and the angle between their ropes is θ . The cart doesn't turn—it moves straight in a direction that results from both pulls happening simultaneously. Using the rule that combines two vectors acting from the same point at an angle, derive an expression for the total force acting on the cart (its magnitude and direction). [3]
- b) The two forces are given by: $\vec{F}_1 = 500 \text{ N, east}$ and $\vec{F}_2 = 600 \text{ N, north}$. Draw a vector diagram to subtract $\vec{F}_2$ from $\vec{F}_1$. Find the magnitude and direction. [2]
14. a) A scalar may be a pure number or a physical quantity. What is the effect of multiplication of a vector by a scalar? [2]
- b) Give the geometrical interpretation of dot product of two vectors. [1]

- c) If $\vec{A} = 4\hat{i} + 6\hat{j} - 3\hat{k}$ and $\vec{B} = -2\hat{i} - 5\hat{j} + 7\hat{k}$, find the angle between $\vec{A}$ and $\vec{B}$. [2]

15. a) The expression of electric field E due to a point charge at any point near to it is defined by $\vec{E} = \lim_{q \rightarrow 0} \frac{\vec{F}}{q}$, where q is the test charge and F is the force acting on it. What is the significance of $\lim_{q \rightarrow 0}$ in this expression? [1]
- b. Why two electric field lines cannot intersect with each other? [1]

- c) Consider three equal charges q_1, q_2 , and $q_3 = q = +1\mu\text{C}$ located at three different points on a circle of radius 1m, as shown in the figure. Calculate the total electric field at point P due to this distribution of charges. [3]



16. a) What are induced and inducing charges. How will you charge a conductor positively by the method of induction. Explain with clear diagram. [1+2]
- b) A small ball of paper has a mass of 9×10^{-5} kg and carries a charge of $5\mu\text{C}$. When it is held over another charged ball of paper at a distance of 2 cm above it, the two balls stay in equilibrium, what is the charge on the second ball? [2]
17. a) What are electric field lines? [1]
- b) If one field line represents 1.0×10^{-9} C of charge, determine the number of field lines drawn for a charge of $2\mu\text{C}$. [1]
- c) Two-point charges $+2\mu\text{C}$ and $+8\mu\text{C}$ are placed 10 cm apart in air. Sketch qualitatively the electric field lines, indicate the region and calculate the position where the resultant field will be zero. [3]
18. a) Write the conditions for validity of Coulomb's law of force between charges. [1]

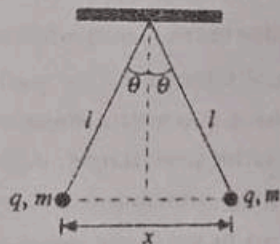
- b) State and explain the principle of superposition of Coulomb's force with suitable diagram. [2]
- c) A polythene piece rubbed with wool is found to have a -ve charge of $0.64 \mu\text{C}$. (i) estimate the number of electrons transferred from which to which? (ii) is there a transfer of mass from wool to polythene? If yes, find the amount of mass transferred. [2]
19. a) Find the dimensions of Planck's constant. If its value in cgs system is $6.62 \times 10^{-27} \text{ erg-sec}$, what will be its value in MKS system? [2]
- b) Assuming that the mass m of the largest stone that can be moved by a flowing river depends upon velocity v , the density ρ and acceleration due to gravity g , show that mass m varies directly as the sixth power of the velocity of the flow. [3]

GROUP 'C' [3×8=24]

Long answer questions:

20. a) What is meant by resolution of a vector? Why resolving a vector is important in physics? Can the walking of a man be an example of resolution of vectors? Explain. [3]
- b) Can the magnitude of the rectangular component of a vector be greater than the magnitude of that vector. Explain. [1]
- c) A man rows a boat with a speed of 18 kmh^{-1} in the north-west direction. The shoreline makes an angle of 15° south of west. Find the components of velocity of the boat along the shoreline and perpendicular to the shoreline. [2]
- d) An athlete runs 2km, east and then 3km, 40° south of east. Find the magnitude and direction of total displacement. [2]
21. a) What do you mean by absolute zero and absolute scale of temperature? [2]
- b) A student claimed that the thermometers register their own temperature therefore, they are useless. How will you respond? [2]

- c) Mercury boils at 357°C . However, mercury thermometers are made such that they can measure temperature up to 500°C . Explain how this is possible. [1]
- d) Suppose you come across old scientific notes that describe a temperature scale called Z on which the boiling point of water is 65.0°Z and the freezing point is -14.0°Z . To what temperature on Fahrenheit scale would a temperature of $T = -98.0^{\circ}\text{Z}$ correspond? Assume that Z scale is linear, that is, size of a Z degree is the same everywhere on the Z-scale. [3]
- a) Explain modern theory of electrification? [1]
- b) Define 1 Coulomb of charge on the basis of Coulomb's law. [1]
- c) When a glass rod is rubbed with a silk cloth, charges appear on both. A similar phenomenon is observed with many other pairs of bodies. Explain how this observation is consistent with the law of conservation of charge. [2]
- d) Two small spheres each of mass m and positive charge q , hang from light threads of lengths l . Each thread makes an angle of θ with the vertical as shown in the figure.



- (i) On the diagram draw and label all the forces on spheres. [1]
- (ii) Develop an expression for the charge q in terms of m , l , θ and g and the Coulomb's law constant, [3]